

Звіт до лабораторної роботи №3

Проект реалізує автоматизований CI/CD для сервісу відстеження завдань з Лабораторії №2.

Пайплайн виконує:

- статичний аналіз
- автоматизоване тестування
- публікацію образів Docker до GHCR
- розгортання через self-hosted runner
- перевірку розгортання

Lint pipeline

Lint pipeline перевіряє:

- Код Java (Checkstyle, PMD, SpotBugs)
- shell скрипти (ShellCheck)
- Dockerfile (Hadolint)
- YAML-файли (Yamllint)

Конвеєр працює на подіях push та pull request.

← Lint

✓ Merge pull request #8 from FruitPug/add-deployment #45

Summary

All jobs

lint

Run details

Usage

Workflow file

Triggered via push 1 hour ago

Status

Total duration

Artifacts

FruitPug pushed - 7ff556a main

Success

41s

-

lint.yml

on: push

lint

38s

Test pipeline

Автоматизовані тести перевіряють ендпоінти застосунку та бізнес-логіку.
JaCoCo використовується для забезпечення мінімального покриття коду 40%.

← Test

✓ Merge pull request #8 from FruitPug/add-deployment #43

Summary

All jobs

✓ test

Run details

Usage

Workflow file

Triggered via push 1 hour ago

FruitPug pushed 7ff556a main

Status

Success

Total duration

31s

Artifacts

1

test.yml

on: push

✓ test

28s

Annotations

1 warning

⚠ test

Node.js 20 actions are deprecated. The following actions are running on Node.js 20 and n
[Show more](#)

Artifacts

Produced during runtime

Name

Size

jacoco-report

74.8 KB

mywebapp

mywebapp

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Ctxty	Missed	Lines	Missed	Methods	Missed	Classes
com.example		66%		56%	9	20	21	72	2	12	1	2
Total	89 of 263	66%	7 of 16	56%	9	20	21	72	2	12	1	2

Pull Request Protection

Правила захисту гілок запобігають мерджу, доки всі перевірки CI не пройдуть успішно.

remove nginx installation #8

Merged FruitPug merged 3 commits into `main` from `add-deployment` 1 hour ago

Conversation 0 Commits 3 Checks 2 Files changed 1

FruitPug commented 1 hour ago Owner ...

No description provided.

FruitPug added 3 commits 1 hour ago

- removed unnecessary nginx installation `b81afb9`
- Merge branch 'add-deployment' of <https://github.com/FruitPug/deployme...> `130d5b4`
- Merge branch 'main' into add-deployment `737d0a2` **Verified**

FruitPug merged commit `7ff556a` into `main` 1 hour ago Hide details Revert

2 checks passed

- ✓ **lint** Details
- ✓ **test** Details

break pipeline intentionally #3

Open FruitPug wants to merge 3 commits into `main` from `broken-pr`

Conversation 0 Commits 3 Checks 2 Files changed 1

FruitPug commented 3 days ago Owner ...

No description provided.

FruitPug added 3 commits 3 days ago

- break pipeline intentionally `6dc7cc9`

All checks have failed
2 failing checks

- ✗ **Lint / lint (pull_request)** Failing after 15s Required ...
- ✗ **Test / test (pull_request)** Failing after 11s Required ...

This branch is out-of-date with the base branch Update branch
Merge the latest changes from main into this branch. This merge commit will be associated with FruitPug.

Merge pull request ▼ You can also merge this with the command line. [View command line instructions.](#)

Merging is blocked due to failing merge requirements

Still in progress? [Convert to draft](#)

Docker Image Publishing

Продакшн Docker образи автоматично створюються та публікуються в реєстрі контейнерів GitHub.

 **deployment-labs/mywebapp** sha-7ff556a Public Latest

 Install from the command line

[Learn more about packages](#)

```
$ docker pull ghcr.io/fruitpug/deployment-labs/mywebapp:sha-7ff556a
```

Recent tagged image versions

latest sha-7ff556a

Published about 2 hours ago · Digest ...

↓ 1

sha-310ac35 v2.0.0 stable

Published about 2 hours ago · Digest ...

↓ 2

sha-9a1e90f

Published about 2 hours ago · Digest ...

↓ 1

sha-5cb6aee v1.1.2

Published about 18 hours ago · Digest ...

↓ 2

← Build and Publish

✓ Merge pull request #8 from FruitPug/add-deployment #10

Summary

All jobs

✓ build

Run details

🕒 Usage

📄 Workflow file

Triggered via push 1 hour ago

Status

Total duration

Artifacts

🚀 FruitPug pushed -> 7ff556a main

Success

1m 21s

1

build.yml

on: push

✓ build

1m 18s

Self-Hosted Runner

Self-hosted GitHub Actions runner виконує розгортання на target VM через SSH.

Runners

[New self-hosted runner](#)

Host your own runners and customize the environment used to run jobs in your GitHub Actions workflows. [Learn more about self-hosted runners.](#)

Runners	Status
 lab3 self-hosted Linux X64 deployment ubuntu-24	● Offline ...

```
github-runner@lab3:~/actions-runner$ ./run.sh
```

```
✓ Connected to GitHub
```

```
Current runner version: '2.334.0'
```

```
2026-05-21 10:29:17Z: Listening for Jobs
```

```
2026-05-21 10:31:24Z: Running job: deploy
```

```
2026-05-21 10:32:45Z: Job deploy completed with result: Failed
```

```
2026-05-21 10:35:54Z: Running job: deploy
```

```
2026-05-21 10:36:53Z: Job deploy completed with result: Succeeded
```

```
^CExiting...
```

```
Runner listener exit with 0 return code, stop the service, no retry needed.
```

```
Exiting runner...
```

Deployment

Розгортання запускається автоматично для анотованих тегів Git.

Додаток розгортається на target VM за допомогою Docker Compose.

The screenshot shows a GitHub Actions workflow run for the 'deploy' job. The status is 'succeeded 1 hour ago in 59s'. The workflow steps are listed on the left, and the 'Annotations' section on the right shows a warning. The steps are:

- Set up job (1s)
- Checkout repository (2s)
- Setup SSH key (0s)
- Copy files to target (3s)
- Run ***ment (47s)
- Verify ***ment (1s)
- Post Checkout repository (1s)
- Complete job (0s)

```
deploy@Ubuntu-server-clean:~$ docker ps
CONTAINER ID   IMAGE                                COMMAND
NAMES
b0133b89a77d   nginx:1.28.3                        "/docker-entrypoint..."
tcp   mywebapp-nginx
bad81ce47cc5   ghcr.io/fruitpug/deployment-labs/mywebapp:latest   "java -jar app.jar"
mywebapp-app
11f637c78ecc   mariadb:12                          "docker-entrypoint.s..."
mywebapp-db
deploy@Ubuntu-server-clean:~$
```

The screenshot shows a web browser window with the address bar displaying '192.168.56.104/'. The page title is 'YAMLLint - The YAML Validator'. The main content area displays 'MyWebApp' in a large, bold font. Below it, the text 'Available endpoint:' is followed by a list of endpoints:

- GET /tasks
- POST /tasks
- GET /tasks/{id}/done

Verification

Перевірка розгортання виявляє неправильні розгортання та автоматично призводить до збою пайплайну.

✓ Verify ***ment

```
1  ► Run ssh \
6
7  nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
8  nginx: configuration file /etc/nginx/nginx.conf test is successful
9  Checking nginx config...
10 % Total    % Received % Xferd  Average Speed   Time    Time       Time  Current
11                               Dload  Upload  Total  Spent  Left  Speed
12
13   0     0    0     0     0     0      0     0  --:--:--  --:--:--  --:--:--    0
14 100     2    0     2     0     0     11     0  --:--:--  --:--:--  --:--:--   11
15 Checking alive endpoint...OK
16 % Total    % Received % Xferd  Average Speed   Time    Time       Time  Current
17                               Dload  Upload  Total  Spent  Left  Speed
18
19   0     0    0     0     0     0      0     0  --:--:--  --:--:--  --:--:--    0
20   0     0    0     0     0     0      0     0  --:--:--  --:--:--  --:--:--    0
21 Checking ready endpoint...OK
22 100     2    0     2     0     0      8     0  --:--:--  --:--:--  --:--:--    8
23 % Total    % Received % Xferd  Average Speed   Time    Time       Time  Current
24                               Dload  Upload  Total  Spent  Left  Speed
25
26   0     0    0     0     0     0      0     0  --:--:~  --:~:~  --:~:~    0
27 100    79    0    79     0     0   1490     0  --:~:~  --:~:~  --:~:~   1519
28 Checking tasks endpoint...[{"id":1,"title":"test","status":"pending","created_at":"2026-05-20 17:06:08"}]
29 Verification successful
```

✗ Verify ***ment

```
1  ► Run ssh \
6
7  nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
8  nginx: configuration file /etc/nginx/nginx.conf test is successful
9  Checking nginx config...
10 % Total    % Received % Xferd  Average Speed   Time    Time       Time  Current
11                               Dload  Upload  Total  Spent  Left  Speed
13   0     0    0     0     0     0      0     0  --:~:~  --:~:~  --:~:~    0
14 Checking alive endpoint...
15   0     0    0     0     0     0      0     0  --:~:~  --:~:~  --:~:~    0
16 curl: (22) The requested URL returned error: 404
17 Error: Process completed with exit code 22.
```

systemd Service

Контейнери Docker Compose керуються через окремий модуль systemd.

```
deploy@Ubuntu-server-clean:~$ sudo systemctl status mywebapp.service
[sudo] password for deploy:
• mywebapp.service - MyWebApp Container Stack
   Loaded: loaded (/etc/systemd/system/mywebapp.service; enabled; preset: enabled)
   Active: active (exited) since Thu 2026-05-21 12:40:02 UTC; 25s ago
     Process: 1425 ExecStart=/usr/bin/docker compose up -d (code=exited, status=0/SUCCESS)
    Main PID: 1425 (code=exited, status=0/SUCCESS)
       CPU: 258ms

May 21 12:39:57 Ubuntu-server-clean systemd[1]: Starting mywebapp.service - MyWebApp Container Stack...
May 21 12:39:58 Ubuntu-server-clean docker[1472]: Container mywebapp-db Running
May 21 12:39:58 Ubuntu-server-clean docker[1472]: Container mywebapp-app Running
May 21 12:39:58 Ubuntu-server-clean docker[1472]: Container mywebapp-nginx Running
May 21 12:39:59 Ubuntu-server-clean docker[1472]: Container mywebapp-db Waiting
May 21 12:40:02 Ubuntu-server-clean docker[1472]: Container mywebapp-db Healthy
May 21 12:40:02 Ubuntu-server-clean systemd[1]: Finished mywebapp.service - MyWebApp Container Stack.
deploy@Ubuntu-server-clean:~$
```